Roll No.: ....

END SEMESTER EXAMINATION DECEMBER 2025

Name of the Program: B.Tech. CSE — Semester: V

Name of the Course: Foundation of Quantum Computing — Course Code: TCS 584

Time: 3 Hours — **Maximum Marks: 100**

Note:

i. **All questions are compulsory.**
ii. Answer any two sub questions among a, b & c in each main questions
iii. Total marks in each main question are twenty.
iv. Each questions carries 20 marks

Q1	(20Marks)	
(a)	Explain the significance of Shor's algorithm and Grover's algorithm in the development of quantum computing.	CO1
(b)	Give one example each of a classical logic gate and its quantum equivalent with proper explanation.	
(c)	Which postulate ensures that quantum evolution is reversible? Explain using unitary operators.	
Q2		
(a)	Explain why quantum evolution must be represented by unitary operators.	CO2
(b)	Define a density matrix. How does it differ from a pure state representation?	
(c)	Discuss how the no-cloning theorem affects quantum communication protocols like BB84 and quantum teleportation.	
Q3		
(a)	What is the difference between product states and entangled states? Give one example of each with suitable explanation.	CO3
(b)	Design a small quantum circuit using CNOT and Hadamard gates that performs the operation of generating a Bell pair.	
(c)	What is quantum parallelism and how does it differ from classical parallel processing?	
Q4		
(a)	For $n = 2$, a balanced function satisfies $f(00) = 0, f(01) = 1, f(10) = 0, f(11) = 1$. Find the amplitude of the $\vert 00\rangle$ term in the final state before measurement.	CO4
(b)	Classically, to determine the secret string s for n input bits, how many oracle queries are required in the worst case? How many are required in the quantum Simon's algorithm?	
(c)	Explain how Grover's algorithm can be used to search for a specific item in an unsorted database of $N = 2^n$. How many oracle queries are required compared to a classical search?	CO5
Q5		
(a)	If each physical qubit in a 7-qubit Steane code has an independent error probability $p = 0.02$, calculate the approximate probability that more than one error occurs (making correction fail).	CO5
(b)	Explain the working principle of the BB84 Quantum Key Distribution protocol. What are the four polarization states used in the protocol?	
(c)	How can Grover's algorithm be applied for database search within ML tasks such as nearest-neighbor search or data clustering?	CO6